Environmental Science and Pollution Research
https://doi.org/10.1007/s11356-023-29770-5

RESEARCH ARTICLE

A predatory social wasp does not avoid nestmates contaminated with a fungal biopesticide

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Received: 3 June 2023 / Accepted: 4 September 2023

Abstract

Fungus-based biopesticides have been used worldwide for crop pest control as a safer alternative to chemical pesticides such as neonicotinoids. Both agrochemicals can be lethal and may also trigger side effects on the behavioral traits of non-target social insects, which play a crucial role in providing essential biological pest control services in agroecosystems. Here, we evaluated whether a commercial formulation of the entomopathogenic fungus *Beauveria bassiana* or the neonicotinoid imidacloprid causes mortality in foragers of *Mischocyttarus metathoracicus*. These social wasps are natural enemies of caterpillars and other herbivorous insects and inhabit both urban and agricultural environments in Brazil. We also tested whether wasps discriminate between biopesticide-exposed and unexposed conspecifics. Through a combination of laboratory (survival assay) and field experiments (lure presentation), along with chemical analyses (cuticular hydrocarbon profiles), we showed that topic exposure to the label rate of each pesticide causes a lethal effect, with the biopesticide exhibiting a slower effect. Moreover, wasps do not discriminate biopesticide-exposed from unexposed conspecifics, likely because of the similarity of their cuticular chemical profiles 24 h after exposure. Overall, the delayed lethal time at the individual level, combined with the indistinctive chemical cues of exposure and the lack of discrimination by conspecifics suggests that the fungal biopesticide may ultimately pose a threat to the colony survival of this predatory wasp.

Keywords Nestmate recognition · Cuticular hydrocarbons · Natural enemy · *Mischocyttarus metathoracicus* · *Beauveria bassiana* · Neonicotinoid

Introduction

In the face of rising demand for food security, large increases in productivity are achieved by agricultural intensification. This has been followed by high international trade that accelerates global dispersion of crop-destroying organisms (Culliney 2014; Sharma et al. 2017) and contributes to an over-dependency on chemical pest control (Guedes et al. 2017). The extensive use of agrochemicals to protect crops negatively affects non-target beneficial organisms, such as pollinators and natural enemies of agricultural pests (Geiger et al. 2010; Guedes et al. 2016; Carvalho 2017). These beneficial organisms might be exposed to agrochemicals either during their application, by contacting contaminated leaves and soil, or by ingestion of nectar and pollen with pesticide residues. Therefore, novel ways to achieve a more sustainable and productive agriculture are urgently needed.

Responsible Editor: Giovanni Benelli

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Published online: 11 September 2023

Environmental-friendly and efficient biopesticides have been developed and applied for crop pest control as a potentially safer alternative to chemical pesticides, such as the neonicotinoids (Mascarin and Jaronski 2016; van Lenteren et al. 2018; Cappa et al. 2022). The use of pathogenic fungus-based biopesticides, such as *Beauveria bassiana* (Bals.-Criv.) Vuill., grows every year, due to its ability to infect and kill multiple species of insects and mites (Mascarin and Jaronski 2016). Through percutaneous infection, the filamentous fungus *B. bassiana* adheres its hydrophobic spores to the cuticular hydrocarbons (CHCs), germinates, and penetrates the insect cuticle, leading to host death (Shah and Pell 2003; Zimmermann 2007). As *B. bassiana*-based biopesticides are effective against target pests, have fast environmental degradation, and their commercial formulations are relatively cheap, they are widely used in croplands (Sayed and Behle 2017; Subbanna et al. 2019). Although such biopesticides are traditionally considered as environmentally sustainable pest control products, a number of recent studies have found detrimental effects on non-target invertebrates (Bordalo et al. 2020; Dornelas et al. 2021; Cappa et al. 2022). Besides, the release of large quantities of biopesticide in croplands (i.e., inundative application) is the most used strategy to suppress pest populations (Castro et al. 2016; Ausique et al. 2017; Wraight et al. 2021), which means likely exposure of (non-)target organisms to high spore concentrations (i.e., field-recommended concentrations) (Goettel and Jaronsky 1997).

As central-place foragers, social hymenopterans often gather food and nesting resources in agroecosystems, where they provide essential pollination and pest control services (Potts et al. 2016; Brock et al. 2021; Anjos et al. 2022). Lethal effects of fungus-based biopesticides have been mostly reported in honey bees, stingless bees, and bumble bees, and to a lesser extent in ants, solitary aphelinid wasps, and social vespid wasps (Bos et al. 2019; Drobnjaković et al. 2019; Mayorga-Ch et al. 2021; Cappa et al. 2022; Leite et al. 2022; Faita et al. 2023). These products also trigger side effects on the physiological, cognitive, and behavioral traits of social insects, which may ultimately compromise their survival (Cappa et al. 2019, 2022; Carlesso et al. 2020; Erler et al. 2022). In social bees, for instance, *B. bassiana* exposure induces subtle changes in their CHC profiles, which are important social recognition cues (Blomquist and Bagnères 2010). In the well-studied honeybee *Apis mellifera* Linnaeus, *B. bassiana*-exposed foragers are highly accepted by unrelated guards, facilitating pathogen transmission between colonies (Cappa et al. 2019). In contrast, guards of the stingless bee *Tetragonisca angustula* (Latreille) detect and repel nestmate foragers exposed to this biopesticide, preventing access to their nests (Almeida et al. 2022).

Social vespid wasps, like many *Polistes* and *Mischocyttarus* species, prey on a variety of pests, acting as biological control agents mainly in smallholding farms (Medeiros et al. 2019; Prezoto et al. 2019; Southon et al. 2019). Beyond the animal protein, social wasps require water, carbohydrates, and nesting sites (Richter 2000) usually provided by agroecosystems, where they are often found (Somavilla et al. 2016; Oliveira et al. 2017; Schoeninger et al. 2020) and may be exposed to commonly used biopesticides. Indeed, social wasps orally exposed to *B. bassiana* show high mortality rates (Harris et al. 2000; Poidatz et al. 2018; Mayorga-Ch et al. 2021). Moreover, many social wasps rely on cuticular chemical cues to make different kinds of social discrimination (Vander Meer et al. 1998; Cini et al. 2019; Cappa et al. 2020), and alterations of CHCs after exposure to *B. bassiana* could disrupt nestmate recognition, leading to increased acceptance of fungus-exposed non-nestmates on their nests (Cappa et al. 2019) or causing rejection of the exposed nestmates (Almeida et al. 2022). This last behavior could prevent pathogen spread into their colony, but it could also reduce the workforce (Almeida et al. 2022), which might be particularly costly for species with colonies composed of a small number of workers. A third possibility is that wasps do not discriminate biopesticide-exposed and unexposed conspecifics, meaning that potentially infected wasps could enter the colony. The hypothesis that fungal exposure alters CHCs and consequently the social recognition in social wasps remains unexplored.

In this context, we performed a broad study comprising laboratory and field experiments with individual wasps and their colonies. First, we addressed the safety of biopesticides compared to synthetic pesticides on non-target organisms. To achieve this, we tested whether topical exposure to commercial formulations based on *B. bassiana* or the neonicotinoid imidacloprid causes lethality in foragers of the Neotropical predatory social wasp *Mischocyttarus metathoracicus* (de Saussure). The neonicotinoid was used here as a toxic reference standard due to its high toxicity to other social wasps (Batista et al. 2022; Teixeira et al. 2022). Second, we investigated potential subtle effects of the biopesticide by testing whether it alters the cuticular chemical profile of exposed wasps and whether the wasps discriminate between biopesticide-exposed and unexposed conspecifics. Given that these wasps are able to perceive very subtle changes in cuticular odors of conspecifics (de Souza et al. 2023), they are good models to test whether the biopesticide alters social recognition. Of note, our behavioral experiments are the first experimental test of nestmate recognition in *Mischocyttarini* wasps.

Methods

Study place and species

Between January and April 2022, we studied a population of *Mischocyttarus metathoracicus* located in the campus of the Universidade de São Paulo, Ribeirão Preto, São Paulo State, Brazil (21°05′S, 47°50′W, and 531 m above sea level), and its surroundings. New colonies of this species are started by a single inseminated female that produces generations of wasps, resulting in a few dozen adults (de Souza et al. 2022) that increase the colony worker force and foraging activity as the cycle progresses. Nests (Fig. 1) are made of vegetal fiber mixed with salivary secretions and are often attached to natural vegetation and man-made substrates commonly found in both urban and agricultural environments in Brazil (de Souza et al. 2020, 2022; Somavilla et al. 2021). Since colonies can remain active all year round and its larvae are fed progressively (e.g., with caterpillars), these diurnal wasps are important natural enemies of pests.

Fungal biopesticide and neonicotinoid

Boveril® was purchased from Koppert "Brasil Sistemas Biológicos Ltda." This compound is a fungal insecticide of contact action based on the *B. bassiana* strain ESALQ PL63 (minimum of 1×10^8 viable conidia/g), corresponding to 50 g/kg of the active ingredient. Right before starting experiments with the wasps, the viability of conidia was estimated at 94%, as determined by counting the number of colony-forming units (CFUs; Alves and Moraes 1998). Evidence® was purchased from Bayer Crop Science. This is a chemical insecticide based on the neurotoxic neonicotinoid imidacloprid, containing 700 g/kg of the active ingredient (A.I.).

Fig. 1 A typical colony of *Mischocyttarus metathoracicus*. On the right, an adult female feeds a larva inside the nest cell with a piece of caterpillar (green ball). Photo by André Rodrigues de Souza

Wasp exposure to pesticides

The following experimental design simulated a worst-case scenario in which wasp foragers could be acutely exposed to the label rate of pesticides at the time of their application in the field (Goettel and Jaronsky 1997). Foragers ($n = 360$ wasps) were collected with forceps from field post-emergent colonies ($n = 200$ colonies) when they brought food to the nest, between 10 A.M and 2 P.M. In the laboratory, they were subjected to one of four different treatments. A quarter ($n = 90$) of them was exposed to the biopesticide by applying 1 μL of an aqueous solution of Triton® X-100 at 0.01% (v/v) containing Boveril® at the recommended label rate (5 μg A.I./μL [1×10^4 viable conidia/μL]) (hereafter Boveril®). In 90 different wasps, we applied 1 μL of an aqueous solution of Triton® X-100 at 0.01% (v/v) without Boveril® (hereafter control 1). Ninety of them were exposed to the neonicotinoid by applying 1 μL of an aqueous solution containing Evidence® at a recommended label rate (0.1 μg A.I./μL) (hereafter Evidence). In the remaining 90 wasps, we applied 1 μL of water (hereafter control 2). Each wasp was isolated in a plastic container of 0.2 L and provided with water and sugar cubes ad libitum (de Souza et al. 2021). Laboratory conditions were 28 °C, 70% of relative humidity, and a light cycle of 12 h light and 12 h dark.

A subset of wasps from each treatment was used in a survival assay (200 wasps, 50 per treatment, each from a different colony). A new subset of the Boveril® and control 1 wasps was killed by freezing 24 h after exposure to be used for chemical analyses (48 wasps: 25 Boveril® and 23 control 1; each wasp came from a different colony) or as lures in the behavioral assay (112 wasps: 56 Boveril® and 56 control 1; both treatments equally balanced within each colony; four wasps per colony, 28 colonies). In the stingless bee *T. angustula*, a period of 2 h was sufficient for the fungus to induce detectable changes in CHCs (Almeida et al. 2022) and affect the bee's response towards exposed conspecifics during social interactions on the nest. Moreover, in *M. metathoracicus*, subtle but detectable changes in CHCs occur 24 h after immune activation by cuticular wounding, which in turn affects social recognition (de Souza et al. 2023).

Survival assay

To test whether exposure to Boveril® or Evidence® causes wasp mortality, we checked the survival of wasps from the different treatments daily for 1 month. They were considered dead when not moving in response to mechanical stimulation (Teixeira et al. 2022).

Chemical analyses

To test whether exposure to Boveril® changes wasp CHCs, we compared Boveril® and control 1 wasps. CHCs were extracted by washing each dead wasp for 1 min (Oi et al. 2021) in a glass vial (Vetec, HPLC) with hexane (Macron Fine Chemicals, 95% *n*-hexane). The glass vials with the extracts were dried in a flow chamber, and the remaining compounds were resuspended in 205 µL of hexane. These compounds were then analyzed by injecting 1 µL of extract (splitless mode) on a gas chromatography-mass spectrometer coupled with a Mass Spectrometer System (GC–MS; Shimadzu, model QP2010) and equipped with a silica capillary column Rxi-1 ms (30 m length × 0.25 mm internal diameter × 0.25 µm film thickness), with helium gas flow at a rate of 1 mL min^{-1}. The oven temperature was initialized at 150 °C and was increasing 4 °C/min until it reached 310 °C (maintained for 10 min). The injector, detector, and transfer line temperature were programmed to be at 250 °C, 250 °C, and 280 °C, respectively. Mass spectrometer scan parameters included electron impact ionization voltage of 70 eV in the 40 to 500 m/z mass range and with a sweeping range of 0.3 s. Hydrocarbon peaks were identified based on their mass spectra fragmentation patterns (ion and molecular mass) and with the aid of a standard solution composed of different synthetic linear hydrocarbons (C_{21}-C_{40} — Fluka analytical) (Carlson et al. 1998).

Behavioral assays

To test whether wasps discriminate conspecifics exposed to Boveril®, either nestmates or non-nestmates, we performed lure presentation assays under field conditions (Cappa et al. 2019; de Souza et al. 2022). Each focal colony ($n = 28$) was presented with four stimuli in a sequentially balanced order that included two nestmates and two non-nestmate lures either unexposed (control 1) or exposed (Boveril®) to the biopesticide. In each assay, a lure was fixed to a small clip at the top of a 30-cm stick and slowly brought near to the nest so that resident wasps could interact physically with the lure. The presentation lasted 1 min starting from the first physical interaction of a wasp with the lure. Bioassays were performed on the same colony at 15-min intervals between each other. All the assays were carried out on a single day, between 10:00 and 17:00, and were video-recorded. Lures comprised dead individuals killed by freezing for 2 days, then allowed to warm for 1–2 h before presentation. This procedure of lure preparation and presentation has been successfully used to test social recognition abilities in *M. metathoracicus* (de Souza et al. 2022, 2023). Each lure was used only once, and we carried out a total of 112 assays.

The researcher responsible for the lure presentation and the researcher responsible for video recordings and analyses were both blind to the lure membership and Boveril® exposure status. The behavior of wasps towards lures was defined as aggressive when they opened the mandibles and attacked the presented lure by biting, holding between mandibles, and stinging the lure. An aggressive act started when a wasp bit, darted, or stung the body of the lure and finished when the wasp stopped biting, darting, and stinging the body of the lure for at least 0.5 s. The frequency and duration of aggressive acts directed toward each lure were recorded by watching each video in slow motion (0.25 s). The total number of adults present on the nest during each assay was also registered.

Statistical analyses

To test whether exposure to Boveril® or Evidence® increases wasp mortality, we performed a parametric survival analysis with Weibull error distribution. Differences among treatments were assessed with contrast analyses, by gradual simplification of the model and grouping treatment levels that were not significantly different ($p < 0.05$) (Crawley 2012).

To test whether exposure to Boveril® causes changes in CHCs, the relative abundance of each compound in Boveril® and control 1 wasps was subjected to a PERMANOVA analysis (Bray–Curtis distances). Non-metric multidimensional scaling (NMDS) analysis (Bray–Curtis distances) was performed to spatially visualize the chemical data.

To test whether wasps recognize conspecific foragers exposed to Boveril®, we ran a generalized linear mixed model (GLMM) based on Poisson distribution (Cappa et al. 2019). In this GLMM, the frequency of the aggressive acts received by each lure was included as the response variable, while the number of adult wasps present on the nest, colony membership, exposure to Boveril®, and the interaction between colony membership and exposure to Boveril® were entered as explanatory variables. Lure nest origin and focal colony were entered as random effects. We also ran a linear mixed model (LMM) in which the total duration of aggressive acts received by each lure was included as the response variable, while the number of adults present on the nest, colony membership, exposure to Boveril®, and the interaction between colony membership and exposure to Boveril® were entered as explanatory variables. Lure nest origin and focal colony were entered as random effects.

The residuals of all models were checked by visual inspection of residuals vs. fitted plots and Q-Qplots, as well as by performing Shapiro–Wilk's test and Bartlett's test. All statistical analyses were performed using R software (R Core Team 2022) with *vegan* (Oksanen et al. 2019), *tidyverse* (Wickham et al. 2019), *lme4* (Bates et al. 2015), and the *lmerTest* (Kuznetsova et al. 2017) packages. The statistical significance level was set at $p<0.05$.

Results

Survival

Treatments affected wasp survival ($\chi^2=143.4$, $df=3$, $p<0.0001$: Fig. 2). Survival of control wasps was similar ($\chi^2=0.001$, $df=1$, $p=0.93$). Boveril® wasps survived lesser than controls did ($\chi^2=9.02$, $df=1$, $p=0.003$). Evidence® wasps survived much less than controls ($\chi^2=143.1$, $df=1$, $p<0.0001$) and Boveril® wasps ($\chi^2=86.5$, $df=1$, $p<0.0001$). The mean lethal time (LT_{50}) was 19 days for Boveril® wasps, but only 1 day for Evidence® wasps. Thus, exposure to Boveril® or Evidence® increased wasp mortality, and exposure to Evidence® caused higher mortality than exposure to Boveril®.

Chemical odor profile

Exposure to Boveril® did not cause changes in wasp CHCs. In both Boveril® and control 1 wasps, we found 24 peaks of hydrocarbons with chain lengths ranging from 25 to 33 carbons (Tables 1 and 2). These compounds comprised linear alkanes (29.2%), methylated alkanes (41.7%), dimethylated alkanes (8.3%), and alkenes (20.8%). Wasps from both treatments had similar CHC profiles (PERMANOVA: $F_{1,46}=1.07$, $p=0.31$) (Fig. 3; Table 1).

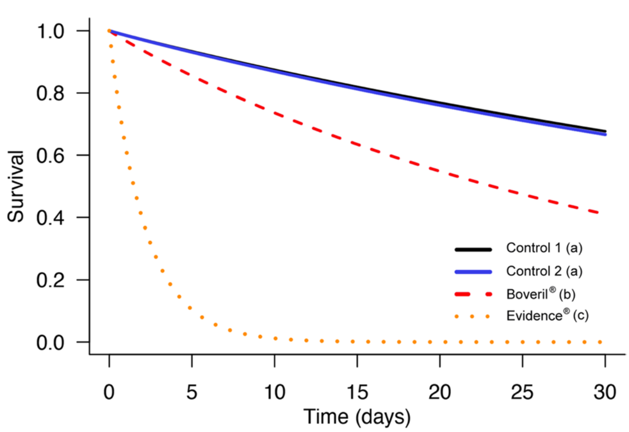

Fig. 2 Survival curves of *Mischocyttarus metathoracicus* foragers over 30 days from an topical exposure in the thorax to 1 µL of an aqueous solution of Triton® X-100 at 0.01% (v/v) containing Boveril® at the recommended label rate (5 µg A.I./µL [1×10^4 viable conidia/µL]) (Boveril®); 1 µL of an aqueous solution of Triton® X-100 at 0.01% (v/v) without Boveril® (Control 1); 1 µL of an aqueous solution containing Evidence® at a recommended label rate (0.1 µg A.I./µL) (Evidence®); 1 µL of water (Control 2). For each treatment, we used 50 wasps, each from a different colony. The curves represent the survival estimate based on a parametric model with Weibull error distribution. Curves coded with different lower-case letters indicate significant differences, based on comparisons performed by gradual simplification of the Weibull model ($p<0.05$)

Social recognition

The duration of aggressive acts received by the lures was affected by colony membership (LMM: $t=2.397$, $df=40$, $p=0.021$) and the number of adult wasps present on the nests (LMM: $t=3.376$, $df=28$, $p=0.002$), but it was not affected by the exposure to Boveril® (LMM: $t=0.288$, $df=50$, $p=0.774$), nor the interaction between nest membership and the exposure to Boveril® (LMM: $t=0.898$, $df=40$, $p=0.375$). The frequency of aggressive acts received by the lures was affected by colony membership (GLMM: $z=6.814$, $p<0.001$), but it was not affected by the exposure to Boveril® (GLMM: $z=0.142$, $p=0.887$), nor the number of adult wasps present on the

Table 1 Hydrocarbons, their retention time, and their relative abundance (mean ± SD) in the cuticle of *Mischocyttarus metathoracicus* wasp according to the exposure to Boveril® at the recommended label rate (5 µg A.I./µL [1×10^4 viable conidia/µL]) (Boveril®), or 1 µL of an aqueous solution of Triton® X-100 at 0.01% (v/v) without Boveril® (Control 1). We used 25 wasps in Boveril® group, and 23 wasps in the control group, each from a different colony

Compounds	Retention time (min)	Control 1	Boveril®
n-C_{25}	23.152	1.373 ± 1.323	1.335 ± 1.070
(Z)-C_{26}	24.554	0.346 ± 0.094	0.406 ± 0.156
n-C_{26}	25.021	0.318 ± 0.101	0.345 ± 0.077
(Z)-C_{27}	26.473	48.471 ± 4.817	46.640 ± 4.547
n-C_{27}	26.851	10.973 ± 2.517	11.419 ± 1.884
13-; 11-; 9MeC_{27}	27.419	0.416 ± 0.251	0.290 ± 0.182
7-MeC_{27}	27.563	0.139 ± 0.062	0.187 ± 0.095
5-MeC_{27}	27.724	0.290 ± 0.070	0.263 ± 0.056
3-MeC_{27}	28.142	11.560 ± 1.899	12.506 ± 1.656
n-C_{28}	28.576	0.410 ± 0.105	0.500 ± 0.146
(Z)-C_{29}-1	29.914	14.835 ± 3.019	14.343 ± 2.254
(Z)-C_{29}-2	30.028	0.906 ± 0.227	0.834 ± 0.164
n-C_{29}	30.275	2.560 ± 0.954	3.193 ± 0.954
15-; 13-; 11-; 9-MeC_{29}	30.795	0.782 ± 0.826	0.547 ± 0.568
7-MeC_{29}	30.956	0.145 ± 0.034	0.138 ± 0.031
7,11-diMeC_{29}	31.420	0.088 ± 0.074	0.098 ± 0.063
3-MeC_{29}	31.496	2.387 ± 0.618	2.872 ± 0.841
n-C_{30}	31.913	0.132 ± 0.075	0.134 ± 0.065
(Z)-C_{31}	33.169	0.433 ± 0.169	0.506 ± 0.221
n-C_{31}	33.500	0.332 ± 0.247	0.439 ± 0.217
15-;13-;11-MeC_{31}	33.973	1.824 ± 2.567	1.407 ± 2.199
3-MeC_{31}	34.652	0.341 ± 0.234	0.592 ± 0.381
17-;15-;13-;11-MeC_{33}	36.965	0.791 ± 0.747	0.805 ± 0.590
11,15-diMeC_{33}	37.358	0.150 ± 0.087	0.204 ± 0.108

Table 2 Results from mixed models on the duration and frequency of aggressive acts received by the lures. *RV* response variable, *EB* exposure to Boveril®, *NM* colony membership, *AW* number of adult wasps on the nest, *LN* lure nest origin, *FC* focal colony

RV	Fixed effects			Random effects	
	t value	*df*	*p*	Variance (SD)	
Duration of aggressions	Intercept: 3.186	Intercept: 53.7528	Intercept: 0.002*	LN	FC
	EB: 0.288	EB: 50.5758	EB: 0.774	Intercept: 0.393 (0.63)	Intercept: 21.141 (4.60)
	NM: 2.397	NM: 39.6506	NM: 0.021*	EB: 146.799 (12.12)	EB: 136.555 (5.14)
	EB/NM: – 0.898	EB/NM: 40.3738	EB/NM: 0.375		
	AW: 3.376	AW: 28.1459	AW: 0.002*		
	z value		*p*		
Number of aggressions	Intercept: 7.338		Intercept: < 0.001*	LN	FC
	EB: – 0.142		EB: 0.887	Intercept: 0.324 (0.57)	Intercept: 0.330 (0.574)
	NM: 6.814		NM: < 0.001*	EB: 0.859 (0.92)	EB: 0.453 (0.67)
	EB/NM: – 1.611		EB/NM: 0.107		
	WN: 1.669		WN: 0.095		

*Statistical difference ($p < 0.05$)

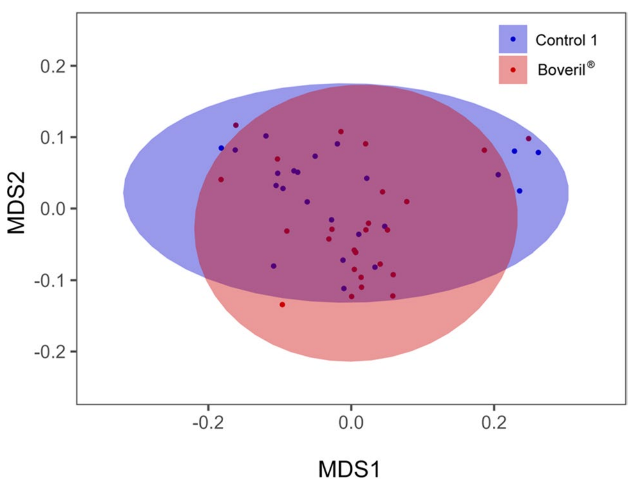

Fig. 3 Non-metric multidimensional scaling (NMDS) analysis of hydrocarbon profiles in the cuticle of *Mischocyttarus metathoracicus* foragers 24 h after topical exposure in the thorax to 1 μL of an aqueous solution of Triton® X-100 at 0.01% (v/v) containing Boveril® at the recommended label rate (5 μg A.I./μL [1×10^4 viable conidia/μL]) (Boveril®, in red), or 1 μL of an aqueous solution of Triton® X-100 at 0.01% (v/v) without Boveril® (Control 1, in blue). We used 25 wasps in Boveril® group, and 23 wasps in the control 1 group, each from a different colony. There was no statistical difference between the groups ($p = 0.31$). Stress value = 0.07

nests (GLMM: $z = 1.669$, $p = 0.095$), nor the interaction between membership and the exposure to Boveril® (GLMM: $z = 1.611$, $p = 0.107$). Thus, non-nestmates received higher levels of aggression than nestmates, and such nestmate recognition was not affected by exposure to the biopesticide (Fig. 4). Wasps did not discriminate exposed from unexposed conspecifics (Fig. 4).

Discussion

We confirmed the hypothesis that a fungus-based biopesticide and a neonicotinoid harm a predatory social wasp. Commercial formulations based on the entomopathogenic fungus *B. bassiana* or the insecticide imidacloprid cause mortality in *M. metathoracicus*, which is delayed in the former. Besides, wasps exposed to the biopesticide (i.e., potentially infected) have CHC profiles indistinguishable from that of unexposed wasps (24 h following exposure) and are not discriminated by conspecifics during experimentally simulated social interactions on nests.

Despite the biopesticide caused significant wasp mortality, all exposed wasps survived the first 24 h, and the LT_{50} for treated wasps was 19 days. As social wasps return daily to their natal colonies, we can speculate that exposed foragers would probably return to the colonies after contamination. In the field, our lure presentation assay simulated social interactions in the nest (de Souza et al. 2022, 2023) and, regardless of the exposure to the biopesticide, *M. metathoracicus* wasps were more aggressive toward conspecific non-nestmates rather than nestmates. To our knowledge, this is the first experimental demonstration of nestmate recognition in *Mischocyttarini*. Our findings demonstrate that the experiment here described was effective in testing recognition abilities even using dead wasps as lures because wasps were able to discriminate nestmates from non-nestmates, following previous studies with similar experimental designs performed with other paper wasps (Signorotti et al. 2014; Baracchi et al. 2015; Cappa et al. 2016a). Moreover, our data show that the exposure to Boveril® does not affect social recognition in *M. metathoracicus*, which means that this biopesticide seems not to trigger acceptance of non-nestmates nor rejection of nestmates, two side

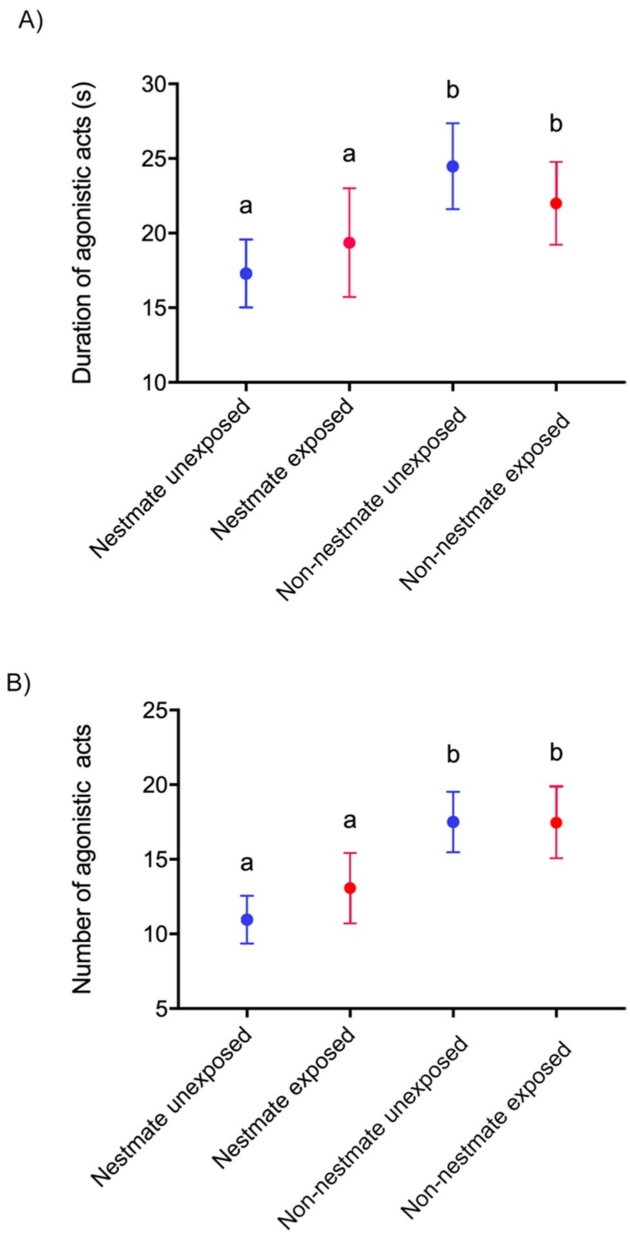

Fig. 4 Duration (**A**) and number (**B**) of aggressive acts (mean ± SE) received by *Mischocyttarus metathoracicus* wasp lures (nestmates or non-nestmates) 24 h after topical exposure in the thorax to 1 µL of an aqueous solution of Triton® X-100 at 0.01% (v/v) containing Boveril® at the recommended label rate (5 µg A.I./µL [1×10^4 viable conidia/µL]) (Boveril®, exposed, in red), or 1 µL of an aqueous solution of Triton® X-100 at 0.01% (v/v) without Boveril® (Control 1, unexposed, in blue). Different letters mean statistical differences between classes ($p < 0.05$)

effects previously reported for social bees exposed to *B. bassiana* (Cappa et al. 2019; Almeida et al. 2022).

Nevertheless, our results show that biopesticide-exposed nestmates are not discriminated by colony mates, which could lead to fungal spread among colony members (Cremer et al. 2007). Horizontal transmission of *B. bassiana* spores from exposed adults to unexposed adults and larvae in small arenas has been demonstrated in the social wasp *Vespula vulgaris* (Linnaeus) (Harris et al. 2000). Therefore, the lethal effect of the biopesticide to directly exposed foragers demonstrated here could potentially extend to their adult and immature nestmates (i.e., indirect exposure), thus providing additional exposure routes that could cause detrimental effects. Consistent with this, Mayorga-Ch et al. (2021) demonstrated dramatic larval mortality in colonies of the social wasp *Polistes myersi* Bequaert, 1934 (Hymenoptera: Vespidae) that fed on *B. bassiana* contaminated prey. This mortality could be attributed to the spread of fungus to the larvae and the consequent infection, or to their intoxication with catabolites produced by the interaction of *B. bassiana* and the wasp's prey, the *Galleria mellonella* (Lepidoptera: Pyralidae) larvae. Also, an analogous effect was reported for *Polistes versicolor* (Olivier, 1791), a paper wasp with a social organization similar to *M. metathoracicus* (Teixeira et al. 2022), in which oral exposure of wasps to azadirachtin significantly impairs colony reproduction (Teixeira et al. 2022), as this biopesticide interferes with endocrine system functions in insects (Sayah et al. 1996).

In contrast to our findings, within 2 h following topic exposure, *B. bassiana* triggered subtle changes in the CHCs of stingless bees (Almeida et al. 2022) and stronger changes in the CHCs of larvae in some coleopterans and lepidopterans (Lecuona et al. 1991). CHCs are one of the most important classes of compounds mediating chemical communication in social insects (Blomquist and Bagnères 2010). Among their functions, CHCs mediate nestmate recognition, a sophisticated communication system that reduces the risk of colony invasions by conspecifics usurpers and natural enemies, including predators and parasites (Breed 2014). Also, CHCs can be used to recognize infected and sick nestmates, therefore reducing the risk of pathogen transmission to the colony (Richard et al. 2008, 2012; Baracchi et al. 2012; Cappa et al. 2016b; Pull et al. 2018; Beani et al. 2019; Almeida et al. 2022; de Souza et al. 2023). Given that pathogens may impair cognitive functions (Klein et al. 2017), disruptions of social and chemical cues add up stress on healthy-compromised colonies, which are less efficient to defend themselves against infected individuals (Cremer et al. 2007). Our data show that wasps were not able to discriminate between pathogen-exposed and unexposed conspecifics, probably due to the similarity of the CHCs. In the field, this behavior may increase the risk of colony contamination, potentially jeopardizing colony dynamics and survival.

The *B. bassiana* has been shown to alter the cuticular chemical profile of honeybee foragers 3 days after exposure (Cappa et al. 2022). Thus, it is possible that the cuticular chemical profile of *M. metathoracicus* foragers changes only after several days from exposure, especially due to the delayed lethality observed in this study. Clearly, there may be additional ways by which exposure to the biopesticide could affect social recognition in wasps. These possibilities should be addressed in future studies.

Although we found no evidence for host-derived chemical cues (CHCs) associated to the biopesticide exposure, pathogen-derived chemical cues were likely present in exposed wasps. For example, ergosterol is a key fungal membrane component, and recent study demonstrated that ants can detect this compound in the cuticle of fungus-exposed conspecifics, which triggers allogrooming behavior (Stock et al. 2023). Therefore, we propose that in addition to an indistinct CHC profile of fungus-exposed wasps, the apparent inability of wasps to detect pathogen-derived chemical cues (e.g., ergosterol) can also explain why exposed conspecifics are not discriminated. We also note that recognition of host-derived chemical cues might largely depend on time since exposure, as it usually takes time for pathogens to induce changes in host phenotype (Lecuona et al. 1991). Instead, recognition of pathogen-derived chemical cues might largely depend on the pathogen load (e.g., number of fungal spores) and the contact area. Thus, exposing wasps in different ways (e.g., deeping, spraying, ingesting) and assessing chemical cues and social recognition after several days following exposure might reveal sublethal effects that our short-term approach did not detect.

In line with the well-established approach that biopesticides are less toxic to non-target species than agrochemicals (Cappa et al. 2022), we found that exposure to the neonicotinoid caused higher wasp mortality than exposure to the biopesticide, even if the active ingredient of the former was diluted 50-fold relative to the latter. Nevertheless, we provided further evidence that harmful effects of fungus-based biopesticides on natural enemies occur, as recently highlighted other species (social wasps: Mayorga-Ch et al. 2021; parasitoid wasps: Rännbäck et al. 2015; predatory bugs: Meyling and Pell 2006; predatory mites: Seiedy et al. 2015; non-biting midges: Bordalo et al. 2020; planarians: Dornelas et al. 2021; Silva et al. 2022; honeybees, bumblebees, and stingless bees: Leite et al. 2022; Cappa et al. 2022; Faita et al. 2023). So, a comprehensive understanding of lethal and a plethora of sublethal effects posed by biopesticides on natural enemies will be essential to provide useful guidelines for development and implementation of strategies to improve integrated pest management practices.

Conclusion

Two commercial formulations of pesticides widely applied in different crops, a biopesticide based on the fungus *B. bassiana* and a neonicotinoid based on imidacloprid, cause mortality in a predatory social wasp considered an effective biological control agent. The delayed lethal toxicity, the absence of chemical CHC cues of exposure, and the lack of discrimination by conspecifics suggest that the biopesticide can be detrimental not only to directly exposed individuals but potentially to their nestmates as well. Yet, the biopesticide is less harmful to directly exposed wasps than the chemical pesticide, supporting the assumption that biopesticides are less toxic to non-target species than chemical pesticides.

Author contribution Conceptualization: André Rodrigues de Souza; writing the original draft: all authors; data curation: André Rodrigues de Souza; Amanda Prato; funding acquisition: André Rodrigues de Souza; investigation: Amanda Prato, Wilson Franca, Sircio Santos, Luan Dias Lima; statistical analyses: Rodrigo Cupertino Bernardes and Eduardo Fernando Santos.

Funding This study was supported by Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) (grants 2020/14464–2 to A.R.S., 2022/07997–0 to A.P., 2019/08029–4 to W.F., 2021/00984–7 to L.D.L., 2020/06632–2 to E.F.S., and 2018/10996–0 to F.S.N.), Coordenação de Aperfeiçoamento de Pessoal de Nível Superior — Brasil (CAPES) — Finance Code 001 (to S.S.), CAPES Print program (grant 88887.571161/2020–00 to M.A.P.L.), and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) (grants 142285/2018–8 to A.P., and 164743/2020–0 to D.A.A.).

Data availability Data will be made available on request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication All authors consent to publish this manuscript.

Competing interests The authors declare no competing interests.

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